Lucas da Rocha Schwengber

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Education

PhD in Statistics — Aug 2024 - Current
University of California, Berkeley, United States.

MS in Applied Mathematics — Aug 2021 - Jun 2023
Instituto de Matemática Pura e Aplicada, IMPA, Brazil.
Advisor: Roberto Imbuzeiro Oliveira
Area: High-Dimensional Probability, Mathematical Statistics and Machine Learning.

BS in Applied Mathematics (with honors) — Mar 2017 - Jun 2021
Universidade Federal do Rio Grande do Sul, UFRGS, Brazil.

Research Projects

Present
Otto's calculus and posterior local sensitivity — 2025-

Past
Geometric planted matchings beyond the Gaussian model (arXiv:2403.17469, accepted on Random Structures and Algorithms) — 2024-2026

Deep Hashing via Householder Quantization (arXiv:2311.04207) — 2023-2025

Industry Experience

Petrobras — IMPA, Mar 2024 - Jul 2024

- Worked in a project for Petrobras at IMPA's Centro Pi. The project involved investigating the use of physics-informed neural networks to solve inverse problems in geophysics. Gained experience with `pytorch`, `tensorboard` and building machine learning pipelines.

Rede Globo — IMPA, Mar 2022 - Jun 2022

- Worked in a project for Rede Globo at IMPA's Centro Pi. The project involved recommendation systems and natural language processing. Worked with `sklearn`, pre-trained models from Hugging Face, `nltk` and collaborative coding using `github`.

Master Soluções que Conectam — Aug 2020 - Dec 2020

- Helped setting up a data collecting system for a project involving computer vision applied to leather data. Mostly worked with `opencv` and `sklearn`.

Teaching Assistanship	*Statistical Machine Learning (STAT154/254)*	UC Berkeley, Jan 2026 - May 2026
	Game Theory (STAT155)	UC Berkeley, Jun 2025 - Aug 2025
	Probability I (Graduate course)	IMPA, Mar 2024 - Jun 2024
	Programming II (Undergraduate course)	IMPA, Aug 2023 - Dec 2023
	Probability I (Graduate course)	IMPA, Mar 2023 - Jun 2023
	Machine Learning (Graduate course)	IMPA, Jan 2023 - Feb 2023
	Precalculus (Undergraduate course)	UFRGS, May 2018 - Jun 2018
Languages	*Portuguese*	Native
	English	Advanced
Code Skills	*Python*	Advanced
	Fortran, Bash, Git	Intermediate
	R	Basic
Honors and Awards	*Berkeley fellowship*	UC Berkeley, 2024-2025
	PIBIC CNPq scholarship	UFRGS, 2017-2018
	Best presentation in the pure Mathematics thematic session at the XXX Scientific Initiation Meeting	UFRGS, Oct 2018
	PIBIC CNPq scholarship	UFRGS, 2020-2021
Talks	*Probability and Combinatorics seminar at UC Irvine*	Irvine, CA, Mar 2026
Events	*Workshop on Learning and Inference from Structured Data: Universality, Correlations and Beyond*	Italy, Jul 2023
	IMPA 70 Years conference	Brazil, Oct 2022
	XXV Brazilian School of Probability	Brazil, Aug 2022
	XXXIII Scientific Initiation Meeting UFRGS	Brazil, Sep 2021
	XXXII Scientific Initiation Meeting UFRGS	Brazil, Sep 2020
	18th School on Time Series and Econometrics	Brazil, Sep 2019
	XXX Scientific Initiation Meeting UFRGS	Brazil, Oct 2018
Summer Courses	*Combinatorics I*	IMPA, Jan 2024 - Feb 2024
	Concentration Inequalities	FGV-EMAp, Jan 2021 - Feb 2021
	Topological Data Analysis	FGV-EMAp, Jan 2021 - Feb 2021
	Markov Chains	IMPA, Jan 2021 - Feb 2021
	Machine Learning and Statistical Modeling	IME-UFRGS, Jan 2020 - Feb 2020
	Summer Course on Bioinformatics	USP, Feb 2018
Undergraduate Research Experience	*Centrality in Complex Networks*	Aug 2019 - Aug 2021
	Advisor: Silvio Dahmen (UFRGS)	Funded by: PIBIC, CNPq

- Studied classical and new centrality measures in complex networks. Used this knowledge to analyze data from historical networks from *Ecclesiastical History of the English People* by Bede. Formulated a new idea of centrality measure with a scale parameter resulting in a technical report (arXiv:2108.09248).

Random Walks and Electric Networks — Jun 2017 - Jun 2018
Advisor: Ricardo Misturini (UFRGS) — Funded by: PIBIC, CNPq

- Covered in detail most of *Random Walks and Eletric Networks* by P. Doyle and J. Snell and simulated some random walks.